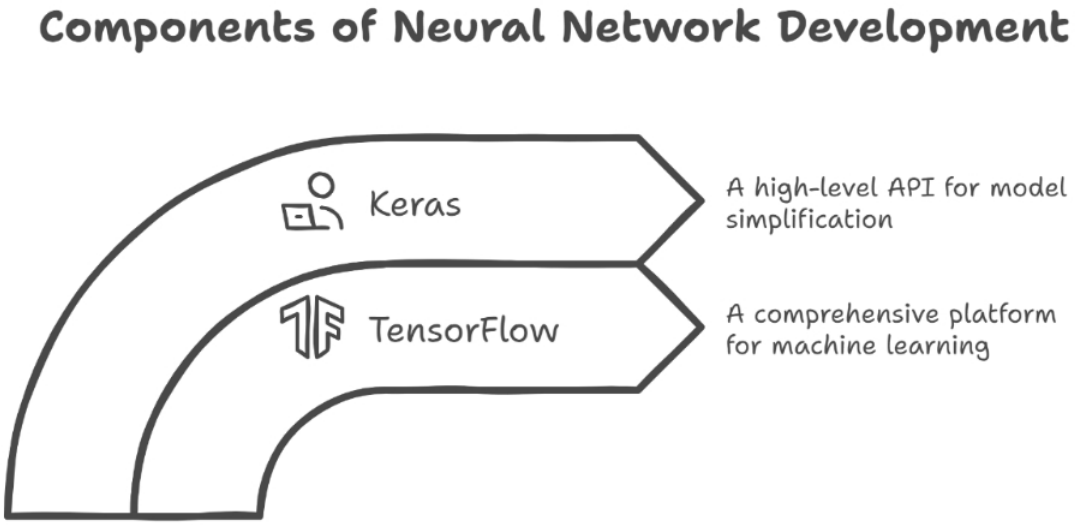
search, and quantum neural networks are paving the way for more efficient and powerful models. Neural networks are poised to play an even more significant role in shaping the future of technology and society.

In summary, neural networks are a transformative technology that mimics the human brain's ability to learn and adapt. By leveraging components such as layers, weights, biases, activation functions, and backpropagation, these networks can tackle some of the most challenging problems in science and industry. With continued advancements, the potential applications and impacts of neural networks are virtually limitless.

Building a Neural Network Using TensorFlow and Keras

TensorFlow and Keras are among the most widely used libraries for developing and training neural networks. TensorFlow is a comprehensive open-source platform for machine learning, while Keras provides a user-friendly, high-level API that simplifies model creation, training, and evaluation.



Step-by-Step Guide to Building a Neural Network

Step 1: Importing Libraries